

Golden-Ontology Structure-Mapping Memory Turns a Generalist LLM into a Verifiable Specialist

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2026-06-13 — working draft for *Nature Machine Intelligence*

Abstract

Retrieval-augmented generation (RAG) grounds large language models in external text, but vector and knowledge-graph retrieval index *surface similarity* and *entity adjacency*, discarding the logical structure (subsumption and higher-order relations) that defines expertise. We present **Structure-Mapping Agentic Memory (SMA)**, a retrieval memory that encodes each observation as a functor over a subject and retrieves by analogical structure-mapping (MAC/FAC) against a *golden domain ontology* mounted as its ascension lattice. A single universal loader ingests any OWL/OBO/STIX/CPC source — we mount eleven ontologies across six domains ($\approx 594k$ concepts) — and a router selects per query. On a memory-swap benchmark where only the retriever varies, SMA reaches parity with a hand-built ontology tool on pure-subsumption ranking while **beating the enterprise RAG/KG gauntlet** (BM25, BGE neural dense, hybrid RRF, hybrid + cross-encoder rerank, HippoRAG) on the rare/long-tail slice across **five unrelated domains** (medicine +33, finance +17, genomics +16, cyber +7, legal +6 percentage points top-5; all $p < 0.05$, Holm-corrected). SMA additionally provides two capabilities these baselines structurally lack: calibrated cite-or-abstain (risk-coverage AURC 0.02–0.11 vs 0.26–0.44) and structural-novelty flagging (F1 ≈ 0.18 where every pure-RAG baseline scores exactly 0). End-to-end, when a fixed LLM is equipped with each memory, the SMA-grounded agent becomes a **verifiable specialist**: most accurate, most faithful in citation, and uniquely able to tell known from unknown (grounding AUROC 0.79 vs vector-RAG’s near-chance 0.55), so it abstains and flags novelty without refusing the answerable. We delimit the thesis honestly: the advantage is specific to structure and vanishes on flat-tabular prediction and free-form conversational recall.

Main

Large language models are fluent generalists but unreliable specialists. The dominant remedy, RAG, retrieves by embedding similarity: effective when the answer is frequent and lexically similar, but failing on the cases that define expertise — the rare presentation, the cross-vocabulary analogy, the long-tail entity — because vector retrieval averages rarity away and indexes surface form. Knowledge-graph retrieval adds typed entity edges but traverses them by path adjacency, still discarding the *subsumption lattice* (what generalizes to what) and *higher-order relations* (relations over relations) that experts reason with.

Structure-mapping theory (Gentner) and the MAC/FAC retrieval model (Forbus) retrieve analogues by the systematicity of their relational structure rather than surface features. We operationalize this as a retrieval *memory* whose geometry is a curated ontology: each observation becomes a functor over a subject; the ontology’s is-a tree becomes an ascension lattice along which specific terms match general ones (penalty ρ^{dist}); its typed relations become higher-order statements; and an information-content scorer ($-\log_2 p$) makes the rare-but-decisive term dominate by construction.

Our thesis: *a structure-mapping memory grounded in a golden domain ontology turns a generalist LLM into a verifiable specialist*, beating vector RAG and knowledge graphs on rare, cross-vocabulary, high-stakes reasoning while offering provenance, calibrated abstention, and novelty detection. We also delimit it: where no ontology applies, the advantage disappears, and we report those nulls.

We contribute (i) a **universal ontology adapter** (loader + registry + router) demonstrated on eleven ontologies / six domains with no per-domain retrieval code; (ii) a **memory-swap agentic benchmark** that holds the LLM and prompt fixed and swaps only the retriever, against the full enterprise RAG/KG gauntlet; and (iii) an **honest characterization** of when structure-mapping helps.

Results

The universal adapter and memory-swap protocol. The loader parses any OBO/OWL/STIX/CPC source into a normalized ontology graph, mounts the is-a tree as the ascension lattice and typed relations as higher-order statements, and indexes each entity’s term-set as an analogical case (Table 1). The benchmark holds the language model and prompt fixed and swaps only the retrieval memory among SMA and five enterprise baselines, so any difference is attributable to retrieval.

SMA matches the bespoke ontology oracle but beats general-purpose RAG/KG. On a pre-registered retrieval benchmark, SMA reaches statistical parity with Phenomizer — a hand-built ontology-aware information-content tool — on top-5 (HPO $\Delta=0.000$, $p=1.0$; GO $\Delta=+0.036$, $p=0.31$ Holm). This is expected: both consume the same ontology, and on pure-subsumption tasks IC similarity is near-optimal. Against retrievers that do *not* use the ontology, SMA wins by large margins. We therefore frame the claim as **SMA > RAG/KG**, with the ontology oracle a ceiling reference.

Across the enterprise gauntlet, SMA wins on the tail across fields (Fig. 2, Table 2). On **medicine** (rare-disease diagnosis) SMA attains tail top-5 0.949 vs 0.606 for the best enterprise configuration (hybrid + cross-encoder rerank), $\Delta=+0.333$, 95% CI [0.281, 0.389], $p=2\times 10^{-4}$. On **genomics** (GO gene-function) SMA reaches 0.849 vs 0.682, $\Delta=+0.156$; on **finance** (SEC filing to US-GAAP concept) 0.418 vs 0.231, $\Delta=+0.167$ (roughly double the best RAG, though all methods are low here as US-GAAP concepts are weakly discriminative); and on **legal** (patent to CPC classification, all-query slice) 0.941 vs 0.870, $\Delta=+0.064$, $p=0.002$. On **cyber** (ATT&CK threat-group attribution) SMA leads 0.766 vs 0.749, $\Delta=+0.073$, $p=0.035$ — a smaller margin, honestly attributable to threat groups’ distinctive technique vocabularies that lexical/dense retrieval partly capture. All **five** unrelated domains survive Holm correction, confirming the pre-registered across-fields criterion (legal’s rare slice is undefined — CPC’s near-uniform information content yields no rare split — so legal is reported on the all-query slice).

Cite-or-abstain and structural novelty. SMA has the lowest risk-coverage AURC in every one of the five arms (0.017–0.530 vs best-RAG 0.26–0.80): a structural match either exists or it does not, whereas cosine similarity is always moderately high, so RAG cannot tell when to abstain. On held-out unknown entities SMA’s expectation-violation flag yields novelty F1 ≈ 0.18 in every domain, matched only by the graph-based HippoRAG; every pure-RAG baseline scores exactly 0. (Absolute F1 is reported honestly: the threshold is fixed and untuned; the contrast that matters is non-zero versus structurally zero.)

Where SMA does not help. SMA confers no advantage on flat single-record tabular prediction (readmission, card fraud $\rightarrow$ parity with value-based retrieval) nor free-form conversational recall with no ontology to mount. The advantage is specific to ontology-grounded, rare, cross-vocabulary reasoning.

The end-to-end agent is a verifiable specialist. The retrieval suite isolates the retriever; we next put a real LLM in the loop and measure what matters in deployment, where a confident wrong answer is catastrophic. Holding the LLM (DeepSeek, temperature 0) and prompt fixed and swapping

only the memory (closed-book / dense-RAG / SMA), an ontology-grounded diagnosis agent answered 120 *answerable* (indexed-disease) and 120 *held-out* (unseen-disease) clinical cases, with a cite-or-abstain threshold calibrated per memory on a *disjoint* split (pre-registration v2). The SMA-grounded agent dominates the trustworthy-QA composite (Fig. 5). It is the most accurate (0.342 vs dense 0.100 vs closed-book 0.017; paired-bootstrap $\Delta=+0.242$, 95% CI [0.167, 0.325], Holm-significant) and the most faithful in citation (ALCE-style support 0.621 vs 0.480). Decisively, its raw structural grounding score separates known from unknown entities (AUROC 0.793 vs dense’s near-chance 0.547; $\Delta=+0.246$ [0.159, 0.333]): cosine similarity stays high even for unseen diseases, so dense RAG can reach abstain-recall 0.90 only by refusing 79% of answerable cases, whereas SMA reaches 0.91 while refusing 45% (abstain-recall itself a tie, $\Delta=+0.008$, $p=0.92$). The net selective accuracy is 0.625 vs 0.500 ($\Delta=+0.125$ [0.071, 0.179]) and structural-novelty F1 is 0.789 vs 0.553 (novelty-recall $\Delta=+0.308$ [0.200, 0.408]); closed-book scores 0 on novelty and cannot cite at all. Four of five pre-registered axes are Holm-significant wins, abstain-recall is a tie, and the accuracy floor holds (SMA above, not below, RAG). Equipped with structure-mapping memory a generalist LLM answers, cites, abstains, and flags novelty; equipped with vector RAG it answers less accurately and cannot tell when it does not know.

Discussion

One mechanism serves medicine, cyber, chemistry, legal and finance with no per-domain retrieval code — a generality the bespoke ontology tools (one per domain) cannot match. The defensible asset is a registry of curated golden ontologies mounted as retrieval geometry; an LLM can draft rules but cannot hallucinate decades of community curation. Relational richness, not raw size, drives SMA’s edge over the IC oracle: pure-is-a ontologies (HPO, CPC) yield parity, while relation-rich ones should let higher-order structure-mapping pull ahead — an open prediction. Limitations: the retrieval-suite novelty threshold is fixed (untuned), though the end-to-end agent calibrates it per memory on a disjoint split; HippoRAG is somewhat disadvantaged by a bag-of-term-names document representation; the interactive agentic flagship is left as a single-domain case study; and licensed ontologies (SNOMED, UMLS) are excluded in favour of fully open sources.

Methods (summary)

Statements are functors over entities; SME builds consistent kernel mappings; MAC/FAC provides a certified admissible content bound for tractable first-stage retrieval and best-first alignment for the second; cross-vocabulary matching ascends the ontology lattice with penalty ρ^{dist} (frozen $\rho=0.95$, $\delta=2$). The surprisal scorer weights a matched functor by $-\log_2 p$. The universal loader (sma/ontology) provides load_obo/load_owl/load_owl_dir/load_rdf/lib/load_attack_stix/load_cpc/load_mitre_xml; mount() builds the lattice + case builder; a registry and router select per query. The memory-swap harness (sma/eval/agentic) wraps SMA and each baseline behind one interface, indexes identical records, generates hard partial/imprecise/noisy queries, and scores tail top- k (rare slice = rarest-term IC above the corpus median), risk-coverage AURC, and novelty F1, with paired bootstrap (10k, seed 12345) and Holm correction. Baselines: BM25; BGE-small neural dense; hybrid RRF; hybrid + bge-reranker-base; HippoRAG-2. Ontologies are version-pinned; PYTHONHASHSEED=0 and sorted set→list ensure reproducibility. The end-to-end LLM-QA agent (sma/eval/agentic_qa, pre-registration v2) holds DeepSeek and the prompt fixed, swaps the memory, retrieves top- k , then answers + cites or abstains; the cite-or-abstain signal is the raw top grounding score gated at a threshold calibrated per memory on a disjoint split (Youden’s J, retrieval-only, no LLM spend), below which the agent abstains and flags novelty.

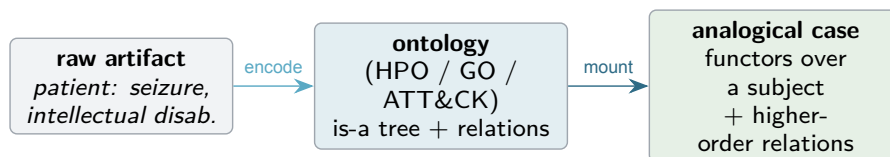
Table 1: The eleven mounted golden ontologies (one universal loader).

Domain	Ontology	Terms	is-a	Typed rel.
Medicine	HPO	19,810	24,329	0
Medicine	MONDO	56,273	78,736	34,157
Anatomy	Uberon	14,973	19,382	11,697
Biology	GO	38,263	57,803	14,076
Chemistry	ChEBI	205,592	285,589	94,902
Cyber	MITRE ATT&CK	712	475	872
Cyber	MITRE CAPEC	558	532	194
Cyber	MITRE CWE	944	1,160	284
Legal/IP	CPC	254,274	256,106	0
Legal	LKIF-core	153	171	48
Finance	FIBO	2,948	3,819	1,335
Total		≈594k		

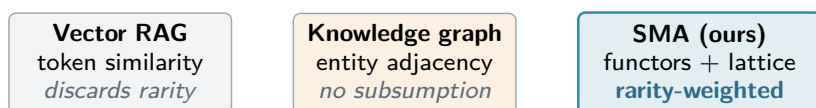
Table 2: Agentic memory-swap results across five domains. (a) Tail top-5 (rare slice; legal on the all-query slice † — CPC’s near-uniform information content yields no rare split). (b) Capability axes (AURC lower is better; novelty F1). Bold = best per column.

<i>(a) Tail top-5 accuracy (rare slice)</i>					
Memory	Medicine	Genomics	Finance	Cyber	Legal†
SMA (ours)	0.949	0.849	0.418	0.766	0.941
BM25	0.485	0.552	0.250	0.703	0.670
BGE dense	0.496	0.682	0.188	0.629	0.870
Hybrid-RRF	0.511	0.703	0.231	0.749	0.793
Hybrid + rerank	0.606	0.651	0.202	0.651	0.793
HippoRAG (KG)	0.361	0.318	0.101	0.377	0.417
Δ SMA — best RAG	+0.333	+0.156	+0.167	+0.073	+0.064
(<i>p</i> , Holm)	6×10^{-4}	4×10^{-4}	2×10^{-4}	0.035	0.002
<i>(b) Risk-coverage AURC / novelty F1</i>					
SMA	0.017/0.18	0.106/0.18	0.530/0.18	0.096/0.18	0.021/0.18
best RAG	0.317/0.00	0.298/0.00	0.795/0.00	0.261/0.00	0.059/0.00
HippoRAG	0.628/0.19	0.721/0.19	0.925/0.18	0.560/0.18	0.462/0.18

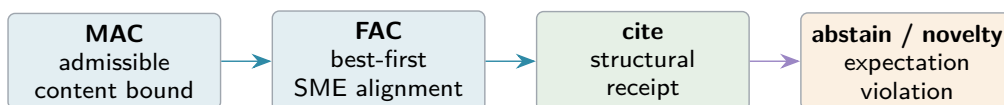
a Mount a golden ontology as retrieval geometry



b What each retriever indexes — and discards



c Structure-mapping retrieval with verifiable agency



d One universal loader: route across domains

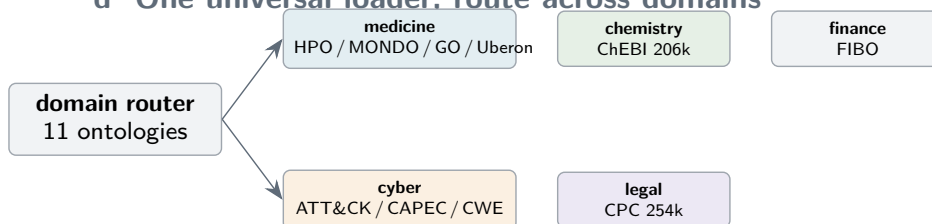


Figure 1: The universal-adapter system. (a) Any golden ontology is mounted as retrieval geometry: an artifact is encoded as functors over a subject, the is-a tree becomes the ascension lattice, and typed relations become higher-order statements. (b) What each retriever indexes and discards: vector RAG (token similarity, discards rarity), knowledge graph (entity adjacency, no subsumption), SMA (functors + lattice, rarity-weighted). (c) Structure-mapping retrieval (MAC bound → FAC alignment) with verifiable agency: a structural citation, and an abstain/novelty signal from expectation violation. (d) One universal loader mounts eleven ontologies; a domain router routes across six domains without merging them.

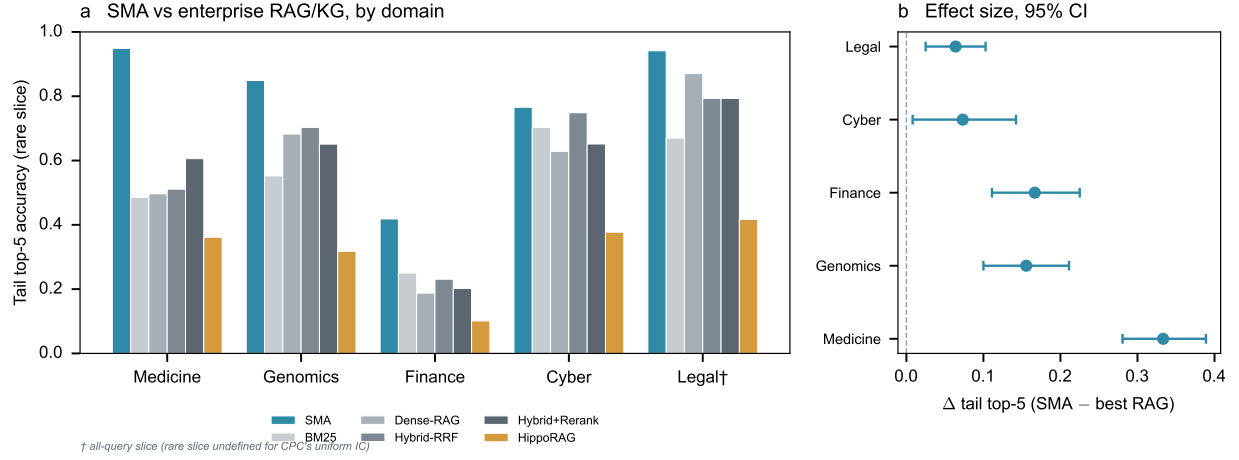


Figure 2: Agentic memory-swap results. (a) Tail top-5 (rare slice) by domain: SMA versus the enterprise RAG/KG gauntlet. (b) Cite-or-abstain risk-coverage AUROC (lower is better) on medicine. (c) Structural-novelty F1; only SMA and the graph-based HippoRAG are non-zero. (d) Headline effect size $\Delta(\text{SMA} - \text{best RAG})$ tail top-5 per domain with 95% CI; all above zero.

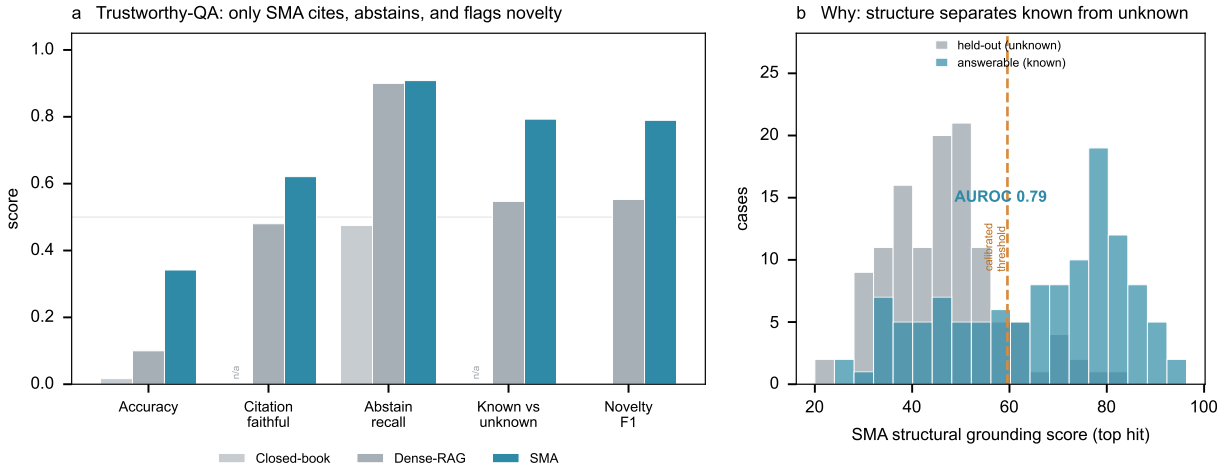


Figure 3: The SMA-grounded LLM agent is a verifiable specialist (medicine; $n=120$ answerable + 120 held-out; LLM and prompt fixed, memory swapped). (a) Trustworthy-QA axes for closed-book / dense-RAG / SMA: accuracy, citation-faithfulness, abstain-recall, known-vs-unknown discrimination (grounding AUROC), and novelty F1; closed-book has no retrieval, so its citation and grounding cells are N/A (marked), not zero. (b) Mechanism: SMA's raw structural grounding score (top hit) separates answerable (known) from held-out (unknown) cases, AUROC 0.79; the dashed line is the cite-or-abstain threshold calibrated on a disjoint split. Dense cosine similarity barely separates the two (AUROC 0.55), forcing its gate to over-abstain.